

Introduction to set

Definition of the set:

A **set** is a collection of objects, things or symbols which are **clearly defined**.

Definition of an elements of the set:

The individual objects in a set are called the **members** or **elements** of the set.

A set must be properly defined so that we can find out whether an object is a member of the set.

1. Listing the elements

The set can be defined by listing all its elements, separated by commas and enclosed within braces. This is called the roster method.

Example:

$$B = \{2, 4, 6, 8, 10\}$$

$$X = \{a, b, c, d, e\}$$

However, in some instances, it may not be possible to list all the elements of a set. In such cases, we could define the set by method 2.

2. Describing the elements

The set can be defined, where possible, by describing the elements. This is called the set-builder notation.

Example:

$$C = \{x: x \text{ is an integer, } x > -3\}$$

This is read as: “C is the set of elements x such that x is an integer greater than -3.”

$$D = \{x: x \text{ is a river in a state}\}$$

We should describe a certain property which all the elements x, in a set, have in common so that we can know whether a particular thing belongs to the set.

We relate a member and a set using the symbol $\in$. If an object x is an element of set A, we write $x \in A$. If an object z is not an element of set A, we write $z \notin A$.

$\in$ denotes “**is an element of**” or “is a member of” or “belongs to”

$\notin$ denotes “**is not an element of**” or “is not a member of” or “does not belong to”

Example:

If $A = \{1, 3, 5\}$ then $1 \in A$ and $2 \notin A$

Note:

1. The empty set or null set is the set that has no elements.
2. The cardinality or cardinal number of a set is the number of elements in a set.
3. Two sets are equivalent if they contain the same number of elements.
4. Two sets are equal if they contain the exact same elements although their order can be different.

Definition and notation used for subsets and proper subsets**Subset:**

If every member of set A is also a member of set B , then A is a subset of B , we write $A \subseteq B$. We can also say A is contained in B .

Proper Subset:

If A is a subset of B , but A is not equal B then A is a proper subset of B , we write $A \subset B$.

The empty set is a subset of any set.

If a set A has n elements that it has 2^n subsets.